$R(A, B, C, D)$ $A \rightarrow C$ (Transitivity)

FD: $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$

S1: Start with all

$(AB\cancel{C}\cancel{D})^+ (A, B, C, D)$

$AD^+ = \{A, B, C, D\}$

$A^+ = \{A, B, C\}$

$D^+ = \{D\}$

$DA = \{B, C, A\}$

$PA = \{A, D\}$

$\alpha = A$

or, $\cancel{A}\cancel{B}\cancel{C}D^+$

$A \to C$

$A \to B$, $B \to C$, $C \to A$

$AB/CD^+ = \{A, B, CD\}$

$A\ D^+ = \{ \ldots \}$

CK so far, AD

$PA = \{A, D\}$

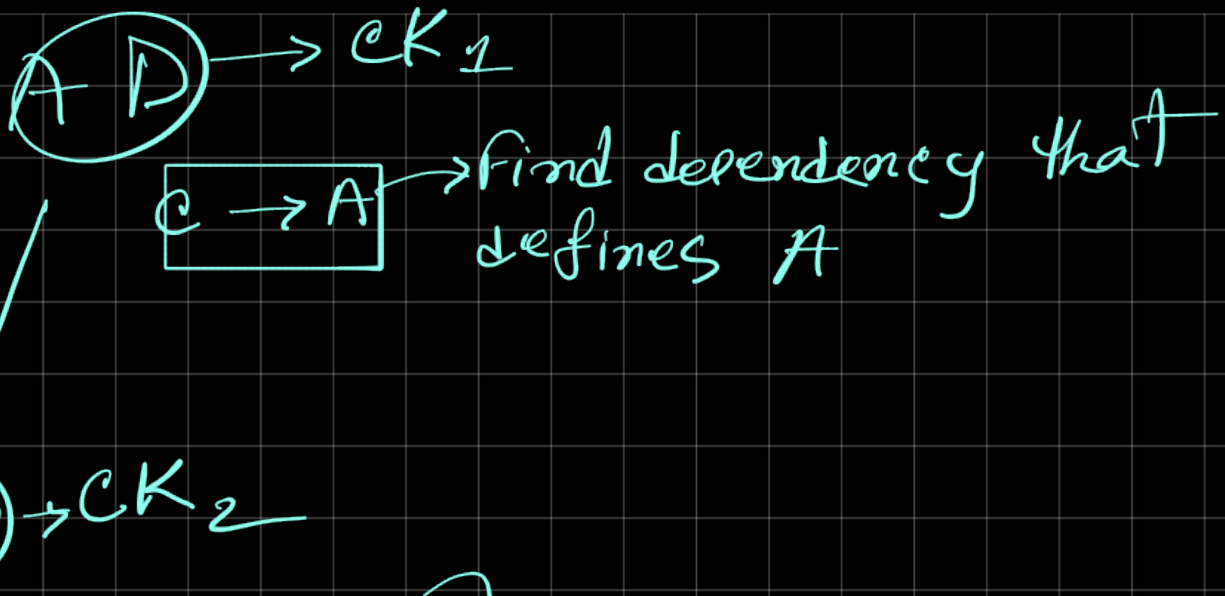
DA → Right HS of part of all Dependencies

$DA = \{B, C, A\}$

$\alpha = PA \cap DA$

$= \alpha$

$x_i \to A$



$C^+ = \{C, A, B\}$

$D^+ = X$

Now, candidate keys discovered so far.

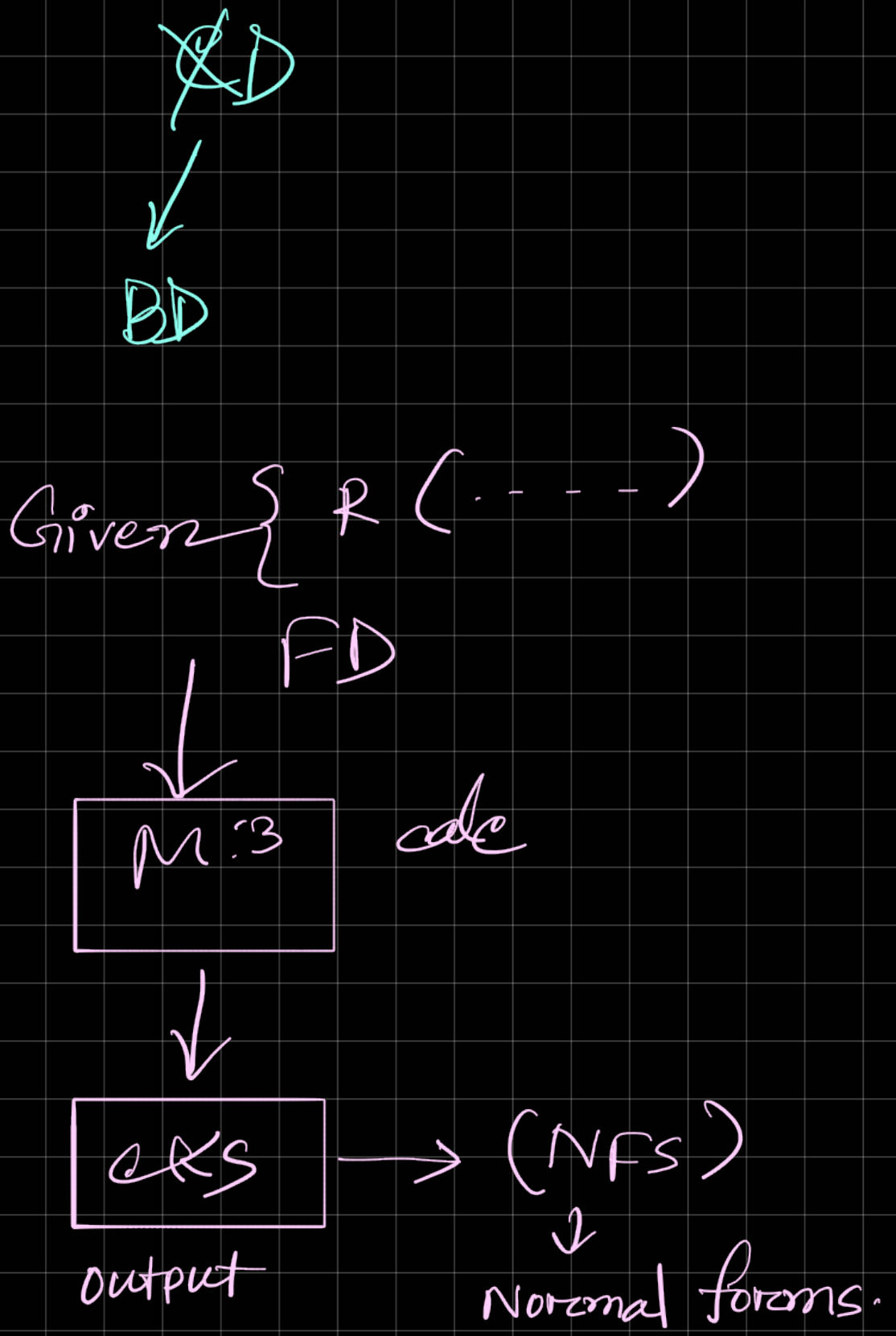
$PA = \{A, D, C\}$ ⟹ CK so far; AD, CD so add to PA

$DA = \{B, C, A\}$

$\alpha = A, C$

So, some $x_i \xrightarrow{\text{can define}} C$

which is B


FD
BD
Given { R (- - - -)
FD
M:3
code
CKS
(NFS)
output
Normal forms.

Normal Forms

↓

To certify whether the design is good / bad.

Atomic Domain

DD → 1NF

each attribute must be atomic.

necessary redundancy.

condition 1

$R(A, B, C, D, E, F)$

FD: $A \to B$ $B \to C$

$C \to D$ $D \to E$

$AB\not{C}\not{D}\not{E}F^+ = \{A, B, C, D, E, F\}$

$AF^+ = \{ \quad - \quad - \quad - \quad \}$

↙

(1) $PA = \{AF\}$

(2) $DA = \{B, C, D, E, \}$

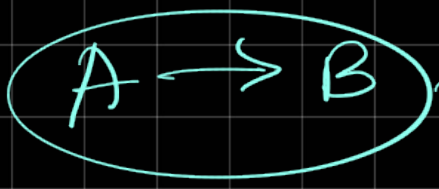
$\alpha = \phi$

Cond 2: proper subset

$NON\,PA = R - (A, F)$

$= B, C, D, E$

CK = AF



→ so voltn has partial dependency so it is not 2NF

R(A, B, C, D)

FD : AB → CD

C → A

D → B

ABCD+ = {A, B, C, D}

AB+ = {all}

A+ = X

B+ = X

$CK_1 = AB$

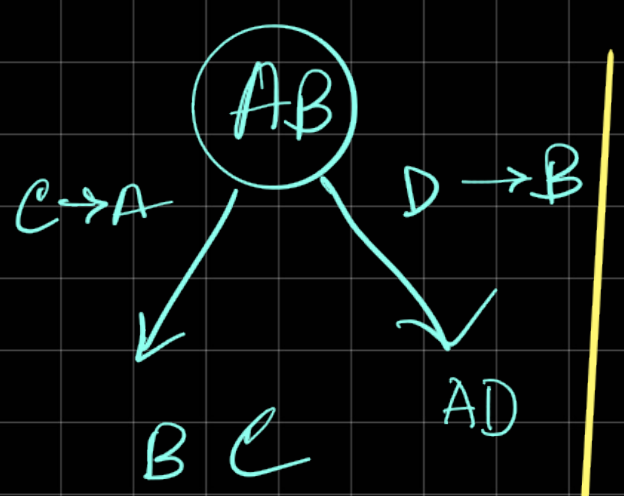
$PA : A, B$

$DA : A, B, C, D$

$\alpha = A, B$

$x_i \longrightarrow A$

$y_i \longrightarrow B$

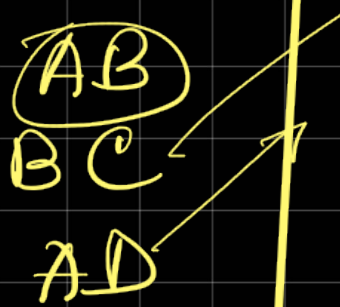


already replaced

$\alpha = AB\,CD$

BC AD

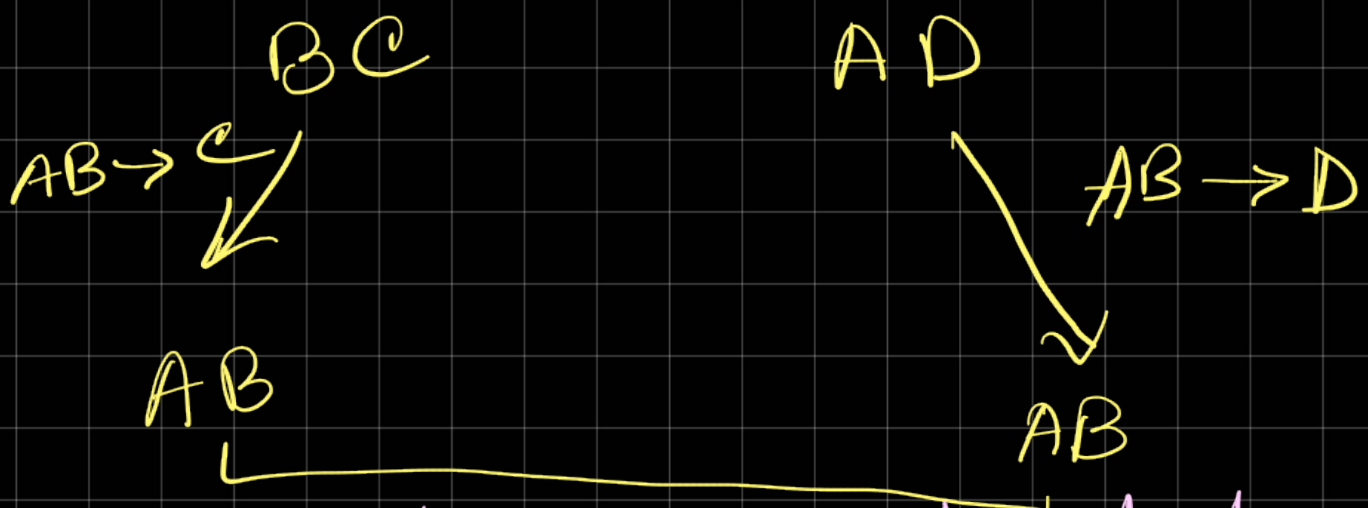
CKS :



$AB \rightarrow CD$

$AB \rightarrow C$

$AB \rightarrow D$



already exists so it ends here.

CKs : AB, BC, AD

Non Prime Attributes

NonPA = R − {A, B, C, D}

= ∅

CKi → NonPA

So in the 2nd Normal Form . 2NF